



User Requirements Specification

Air Con Energy Saving, Monitoring and Management Solution · consolidated from the stakeholder discussion

Document	D5 — User Requirements Specification (URS)
Product	ARUSIQ Smart AC Intelligence Platform
Source	Stakeholder discussion result: identity, multilingual and voice access, three dashboards, split unit first then HVAC, home/office categories, proactive maintenance with red/orange/green signals, payment-linked cooling control, energy savings, air quality, hidden tamper-proof gadget, Digital MRV, tokenized offsets, Scope 2, carbon-platform API
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Relationship to other documents	Consolidates and supersedes section 1 (URS) of the Smart AC Phase 1 Product Definition Pack v0.1. Existing UR IDs are kept; new families UR-MNT, UR-PAY, UR-ENE, UR-AIR, UR-CAR, UR-DEV, UR-HVAC, UR-NFR are added. Satisfied by D6 PRD, D1 Use Cases, D2 UI/UX, D3 Architecture, D4 Rules & Carbon Methodology.

STATUS Draft for stakeholder validation. This document is part of the Phase 1 design set and extends the Smart AC Phase 1 Product Definition Pack (URS, PRD, roles, journeys, sitemap, acceptance criteria). Identifiers UR-, EP-, J0x, SCR- and AC- refer to that pack.

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1. Purpose, scope and conventions

This document states what users and the business need from the solution, in their words, independent of how it will be built. It is the contract against which the product requirements (D6), the design documents (D1 to D4) and the acceptance tests are checked.

1.1 Scope

- **Phase 1 — split-unit air conditioners.** First a clickable UI/UX prototype (Phase 1A), then the functional platform (1B) and the small, low-cost gadget with firmware inserted inside the air conditioner (1C).
- **Phase 2 — HVAC.** Central, cassette and other systems of any brand through gateways, reusing the same applications and rules.
- **Business models covered.** Rent-to-own with payment-linked service control; general maintenance contracts (regular, reactive, preventive); carbon and CO₂ offset as a service with energy savings; other programmes to be inserted later.

1.2 Priority scheme

Priority	Meaning
Must	Required for the phase in which it is listed; the release is not accepted without it.
Should	Important; included unless cost or time forces deferral, in which case the next phase is committed.
Could	Desirable; included when capacity allows.
Future	Recorded so that the design does not prevent it; not committed.

1.3 Definitions

Term	Meaning in this document
Unit	One air conditioner (split wall, cassette, window, central, VRF) with its indoor and outdoor parts.
Gadget / Smart AC Device	The small, affordable, hidden IoT module with sensors and control installed inside the unit, with firmware.
Space	A room or area on a floor of a property, in the category Home or Office.
Signal	A measured or derived value that describes the state of a part or of the room.
Alert	A red, orange or green (and grey for unknown) notification generated from signals, with an action.
Restriction	A governed reduction of cooling (slower, warmer, limited hours) or stop applied by Admin when payments are not made, or by another programme.
Baseline / normal	What the unit would have consumed without the solution, used to show savings.
Digital MRV	Digital monitoring, reporting and verification of energy saved against the baseline, giving audit-ready proof.
CO ₂ ppm vs kgCO ₂ e	Indoor carbon-dioxide concentration (air freshness) versus greenhouse-gas emissions from electricity; never mixed.

2. Stakeholders and user classes

User class	Who	What they need from the solution
End user / Client	Home owner, tenant, office occupant or facility user; may be on rent-to-own, subscription or maintenance contract	A strong, visual, graphical dashboard: comfort, air quality, energy cost and savings, alerts, remote control, service requests, payment status, carbon saved

User class	Who	What they need from the solution
Technician — internal	Company field engineer	A strong visual dashboard of assigned work, prioritised alerts, part-level evidence, checklists and closure
Technician — 3rd party	Partner or contractor engineer	The same, limited strictly to assigned work
Admin / HQ	Operations, service, finance and sustainability staff at headquarters	A strong visual dashboard of the whole fleet, alerts, dispatch, payments and restrictions, energy and carbon reporting, configuration and audit
Business owner	Company management	Commercial control (rent-to-own), maintenance revenue, carbon-offset service, expansion to HVAC and other brands and markets
Verifier / carbon platform	External	Audit-ready evidence packages and an API for verified reductions

3. User requirements

Requirements are grouped by theme. The Source column quotes or paraphrases the discussion item that the requirement comes from. IDs that also exist in the Product Definition Pack keep their number.

3.1 Identity and access (UR-AUTH)

ID	Requirement statement	Priority	Source (discussion)	Phase
UR-AUTH-01	A user can sign in with their credentials (or a seeded demo account in the prototype) and land on the dashboard for their role.	Must	Sign In	1A
UR-AUTH-02	A user can sign out from the current device and, optionally, from all devices.	Must	Sign Out	1A
UR-AUTH-03	A user who forgot their password can recover access by email or WhatsApp without contacting support, with a neutral confirmation and expiry. 1A: Mocked if possible and applicable	Must	Forget Password	1A
UR-AUTH-04	A user sees only the navigation, data and actions permitted for their role and scope (own, assigned, all).	Must	Three role dashboards	1A

3.2 Language and voice (UR-LANG, UR-VOICE)

ID	Requirement statement	Priority	Source (discussion)	Phase
UR-LANG-01	A user can choose the language of the whole product (screens, notifications, reports) from a list; English and Bahasa Indonesia at launch, any further language as a locale pack without code change.	Must	In any language text as the option	1A
UR-LANG-02	Dates, numbers, currency, temperature units and message templates follow the chosen language and locale.	Must	Any language	1A

3.3 Dashboards and asset structure (UR-CLI, UR-TEC, UR-ADM, UR-ASSET)

ID	Requirement statement	Priority	Source (discussion)	Phase
UR-ASSET-01	A user can organise and navigate units by category Home or Office, then property, floor or area, room or space, then unit, and every level shows the worst status of its children.	Must	Select categories: home or office, which area/floor/room or space	1A

ID	Requirement statement	Priority	Source (discussion)	Phase
UR-CLI-01	A client sees a strong visual graphical dashboard summarising comfort, unit health, energy, savings, alerts and maintenance for their spaces.	Must	End users with strong visible and graphical dashboard	1A
UR-CLI-02	A client can open any unit and see its current state, trends and health of indoor, outdoor and electrical parts.	Must	Monitor every aspect of the split air con	1A
UR-CLI-03	A client sees room temperature, humidity and indoor CO ₂ where sensors are available.	Must	Air quality and temperature seen on apps	1A
UR-CLI-04	A client sees an indoor-air-quality interpretation (fresh, stale, humid, dusty) with the contributing data.	Should	Fresh air alerts, health protection	1A
UR-CLI-05	A client sees current energy, estimated cost and actual-versus-normal (baseline) comparison showing the savings with the solution.	Must	Show the savings with our solution and normal what it will be	1A
UR-CLI-06	A client sees estimated Scope 2 emissions and avoided emissions with the method and verification status.	Should	CO ₂ and carbon offset as a service	1A
UR-CLI-07	A client can control authorised settings remotely and create schedules and automations.	Must	Remote control, schedules	1A
UR-CLI-08	A client receives abnormal-condition and maintenance notifications in real time.	Must	Real time notifications	1A
UR-CLI-09	A client can create and track a service request.	Must	Plan and proactively help to manage and maintain	1A
UR-CLI-10	A client can view maintenance history and upcoming service.	Should	General maintenance	1A
UR-CLI-11	A client can view payment status, notices and any grace-period or restriction information, and pay through the linked channels. 1A: Mock if possible and applicable	Must	Payments not made; notified via WhatsApp/email linked to card or payment instructions	1A
UR-TEC-01	A technician sees a strong visual dashboard of assigned work orders with prioritised critical cases and SLA.	Must	Technicians with strong visible and graphical dashboard	1A
UR-TEC-02	A technician can access only assigned customers, sites and units; third-party technicians only assigned work orders.	Must	Internal & 3rd parties	1A
UR-TEC-03	A technician can review current and historical sensor evidence for a unit.	Must	Know what the problem is in advance	1A
UR-TEC-04	A technician sees the suspected fault, its confidence and recommended verification steps.	Must	Predictive diagnostics	1A
UR-TEC-05	A technician can distinguish confirmed, suspected, data-quality and sensor issues.	Must	Predictive maintenance	1A
UR-TEC-06	A technician can complete an inspection checklist per part group and record findings.	Must	Indoor, outdoor, electrical parts	1A
UR-TEC-07	A technician can add notes, photos and spare parts to a work order.	Should	Maintenance	1A
UR-TEC-08	A technician can resolve or escalate a work order with evidence and the client can confirm the result.	Must	Maintenance	1A
UR-TEC-09	A technician sees device offline, power-loss and possible-tamper states.	Must	Alerted if the gadget is removed	1A

ID	Requirement statement	Priority	Source (discussion)	Phase
UR-ADM-01	Admin/HQ sees a strong visual dashboard of fleet-wide health, alerts, connectivity, energy, service backlog and payments.	Must	Admin/HQ with strong visible and graphical dashboard	1A
UR-ADM-02	Admin/HQ can drill down by organisation, property, floor, room and unit.	Must	Categories	1A
UR-ADM-03	Admin/HQ can view and acknowledge critical alerts and dispatch service.	Must	Plan and proactively help	1A
UR-ADM-04	Admin/HQ can manage users, roles and organisation assignments.	Should	Roles	1A
UR-ADM-05	Admin/HQ can view payment status and grace-period cases.	Must	Payments not made	1A
UR-ADM-06	Admin/HQ can review and approve a safe restriction of cooling (slow down or stop) for non-payment.	Must	Cut off or slow down the air con coolness by Admin	1A
UR-ADM-07	Admin/HQ sees the command lifecycle from request through device verification.	Must	Remote control	1A
UR-ADM-08	Admin/HQ can view technician assignments, work-order status and SLA summary.	Must	Maintenance	1A
UR-ADM-09	Admin/HQ can configure or preview alert thresholds and escalation rules.	Should	Signals & notifications	1A
UR-ADM-10	Admin/HQ can view tariff and grid emission-factor settings with effective dates.	Should	Regional grid emission factors	1A
UR-ADM-11	Admin/HQ can review Digital MRV-ready reports and data-quality status.	Should	Digital MRV	1A
UR-ADM-12	Admin/HQ can connect to a carbon platform to exchange or offset verified reductions (mock in the prototype).	Could	Link to a platform to exchange and/or offset	1A
UR-ADM-13	Admin/HQ can review a complete audit trail for sensitive actions and calculations.	Must	Governance of payment control	1A

3.4 Monitoring, signals and maintenance (UR-MNT)

The solution must monitor every aspect of the unit so that problems are known in advance or as they happen, with red, orange and green alerts and a recommended action.

ID	Requirement statement	Priority	Source (discussion)	Phase
UR-MNT-01	Every alert is classified green (normal), orange (warning, plan action), red (critical, act now) or grey (offline or unknown), always with colour, icon and label, and with the evidence and recommended action.	Must	Signals & notifications with red, orange and green alerts	1A
UR-MNT-02	Indoor unit parts are monitored: air filters, evaporator coils, blower motor and fan, condensate drain line and pan, vents and louvers.	Must	Indoor unit parts	1A (simulated) / 1C (live)
UR-MNT-03	Outdoor unit parts are monitored: condenser coils, compressor, condenser fan and blades, refrigerant lines.	Must	Outdoor unit parts	1A / 1C
UR-MNT-04	Electrical and control parts are monitored: thermostat and sensors, capacitor and contactor, electrical wiring.	Must	Electrical and control parts	1A / 1C
UR-MNT-07	The system supports regular, reactive and preventive maintenance workflows for customers not on rent-to-own, including contracts, cadence, visits and history.	Must	General maintenance (regular, reactive and preventive)	1A

ID	Requirement statement	Priority	Source (discussion)	Phase
UR-MNT-08	A predicted problem is labelled suspected until a technician verifies it; the post-service readings confirm the fix.	Must	Know the problem in advance	1A

3.5 Payment-linked service control and other programmes (UR-PAY)

ID	Requirement statement	Priority	Source (discussion)	Phase
UR-PAY-01	Admin can slow down the coldness of a unit or stop it when the user has not paid, for rent-to-own services, through a governed process.	Must	Cut off or slow down by Admin if payments are not made	1A (simulated) / 1B
UR-PAY-03	Restriction follows a ladder (reminder, warmer minimum setpoint, limited hours, stop) with approval and audit at each step; a stop is never applied to health-sensitive spaces.	Must	Slow down or stop	1A / 1B

3.6 Energy savings and automation (UR-ENE)

ID	Requirement statement	Priority	Source (discussion)	Phase
UR-ENE-01	The solution tracks electricity usage of each unit in real time and shows cost from the applicable tariff.	Must	Lower energy bills: track electricity usage in real time	1A / 1C
UR-ENE-02	A user can set schedules so the unit turns off when nobody is home and turns on before they arrive.	Must	Set schedules	1A
UR-ENE-03	The unit turns off automatically when the room is empty (occupancy) and cooling is throttled when users leave and restored on return (geofencing with consent). 1A Stage: Show mock status only	Should	Auto turn off if room is empty; dynamic geofencing	1A
UR-ENE-06	The solution learns the daily routine, adjusts temperatures automatically and syncs with local weather to optimise cooling (automated comfort). 1A: Show mock status	Should	Automated comfort	1B
UR-ENE-07	The solution shows the savings achieved with the solution against what normal operation would have consumed, with the method disclosed; target 10–20 % or more less wasted run time.	Must	Cut wasted run time by 10 % to 20 % or more; show savings vs normal	1A

3.7 Air quality, ventilation and health protection (UR-AIR)

ID	Requirement statement	Priority	Source (discussion)	Phase
UR-AIR-01	The solution measures indoor CO ₂ concentration, humidity and, where fitted, dust (PM2.5) and volatile compounds, and shows them with plain-language bands.	Should	Better air quality; track dust, humidity, allergens	1A / 1C
UR-AIR-02	The solution alerts on stale air (high CO ₂) that causes sleepiness and poor focus and recommends ventilation.	Should	Fresh air alerts	1A / 1C
UR-AIR-04	The solution alerts when filters need washing or replacing to keep air fresh, and tracks filter cleanliness.	Must	Alert when filters need washing or replacing	1A
UR-AIR-06	Indoor CO ₂ ppm is always presented separately from carbon emissions in kgCO ₂ e.	Must	How do you calculate the CO ₂	1A

3.8 Carbon, Digital MRV, offsets and platform integration (UR-CAR)

ID	Requirement statement	Priority	Source (discussion)	Phase
UR-CAR-01	The solution calculates electricity-related emissions as energy used multiplied by a versioned regional grid emission factor (Scope 2, location- and market-based).	Must	Scope 2 emission auditing; regional grid emission factors	1A
UR-CAR-02	The solution calculates avoided emissions from energy saved against the baseline and labels them estimated, provisional or verified.	Must	Energy savings; high-quality carbon credits	1A
UR-CAR-04	Verified reductions can be linked to decentralised-ledger applications or carbon management platforms so property owners can trade accumulated savings as micro-offsets on voluntary markets.	Should	Tokenized carbon offsets	Future (mock in 1A)
UR-CAR-05	A user who wishes can offset their remaining emissions or exchange their verified reductions through a connected platform, with explicit confirmation and a certificate.	Should	Offset carbon if one wishes and link to a platform to exchange and/or offset	Future (mock in 1A)
UR-CAR-06	Energy reductions are converted into formal corporate sustainability reports by tenant, floor, site and period. 1A: Simple report only	Should	Converting smart-home actions into formal corporate sustainability reports	1B
UR-CAR-08	Avoided emissions are never presented as issued carbon credits until independently verified under an accepted methodology.	Must	High-quality carbon credits	1A

3.11 Communications (UR-COM)

ID	Requirement statement	Priority	Source (discussion)	Phase
UR-COM-01	Notifications are delivered in-app and by WhatsApp and/or email according to severity and user preference, in the user's language, with delivery status. 1A: Mock if possible and applicable	Must	WhatsApp and/or email notification	1A
UR-COM-02	Every notification carries the same title as the in-app alert and a link to the relevant screen or payment page. 1A: Mock if possible and applicable	Must	Notifications linked to payment instructions	1A

3.12 System and non-functional requirements (UR-SYS, UR-NFR)

ID	Requirement statement	Priority	Source (discussion)	Phase
UR-SYS-01	In the prototype all telemetry is marked simulated and shows data freshness.	Must	UI/UX clickable first	1A
UR-SYS-02	Offline or unavailable information is represented without fabricating values.	Must	Integrity	1A
UR-SYS-03	The product works on desktop, tablet and mobile, as a web app and an installable app.	Must	Seen on apps	1A
UR-SYS-04	Empty, loading, error and no-data states are realistic.	Should	UX quality	1A
UR-SYS-05	Role switching cannot expose unauthorised data.	Must	Roles	1A
UR-NFR-01	Real-time: a change at the unit is visible in the app within 30 seconds; a critical alert is delivered within 60 seconds.	Must	Real-time notifications	1B
UR-NFR-02	Multi-tenant: many organisations share the platform with strict data separation.	Must	Company, clients, 3rd parties	1B

ID	Requirement statement	Priority	Source (discussion)	Phase
UR-NFR-03	Security and privacy: encryption in transit and at rest, per-device identity, consent for location and voice, compliance with data-protection law in each market.	Must	Hidden gadget, payments	1B
UR-NFR-04	Availability 99.5 % for the cloud platform; devices keep enforcing policies locally when offline.	Should	Reliability	1C
UR-NFR-05	Accessibility: status never colour-only, keyboard and screen-reader support, voice with text alternative.	Must	Strong visible dashboards; voice	1A
UR-NFR-06	Scalability to thousands of units per organisation and multiple countries with local tariffs, grid factors and languages.	Should	Localised markets	1B

4. Constraints and assumptions

- The first deliverable is the clickable UI/UX prototype for split units with simulated data; the gadget and firmware follow in Phase 1C. Sample requirements from phase 1C and 1B is included but will not be the main focus and will not be treated as such, and will be omitted by default in the Phase 1A Prototype in order to meet time constraints.
- Restriction of cooling for non-payment requires contractual consent, legal review per market and a safety policy; the prototype demonstrates the governed process, not instant cut-off.
- A split unit alone recirculates air; fresh-air control needs a connected fan, fresh-air unit or HVAC damper.
- Savings are estimates against a baseline until independently verified; carbon credits need an accepted methodology and a verification body.
- English and Bahasa Indonesia are the initial languages; other languages are locale packs.
- Brand-agnostic control depends on infrared codebooks per brand and, for HVAC, on site protocols and vendor gateways.
- Simulations, estimations, and unverified values are to be considered the default in this clickable prototype. Features will be mocked as needed.

5. Acceptance principles

1. Each Must requirement for a phase should be able to be demonstrated to stakeholders.
2. Simulated, estimated and unverified values are to be disclosed when applicable.
3. No screen is a dead end; every alert leads to an action.
4. Sensitive actions (restriction, carbon transactions, configuration) show request, approval, execution, verification and audit.
5. Role scoping is proven by attempting unauthorised navigation for each role.

6. Traceability

Requirement family	PRD epics (D6)	Use cases (D1)	Design (D2–D4)
UR-AUTH, UR-LANG, UR-VOICE	EP-01, EP-02, EP-13	UC-A01..A05	D2 §3.1, §6
UR-ASSET, UR-CLI, UR-TEC, UR-ADM	EP-02..EP-08, EP-15	UC-B01..B04, UC-C01..C03, UC-E01..E04, UC-H03..H04	D2 §4
UR-MNT	EP-05, EP-06, EP-07, EP-16	UC-C01..C03, UC-E01..E04	D4 §2–3
UR-PAY	EP-09	UC-F01..F03	D3 §6
UR-ENE	EP-08, EP-10, EP-17	UC-B04, UC-D02, UC-D04	D4 §5
UR-AIR	EP-11, EP-18	UC-B03	D4 §4

Requirement family	PRD epics (D6)	Use cases (D1)	Design (D2–D4)
UR-CAR	EP-12, EP-19	UC-G01..G04	D3 §8, D4 §6–8
UR-DEV	EP-14, EP-20	UC-H01, UC-H02	D3 §3
UR-HVAC	EP-21	UC-H02 A1	D3 §12
UR-COM, UR-SYS, UR-NFR	EP-06, EP-22	All	D2 §5, D3 §9